A Diversity Metric for LLM Outputs via Progressive Conditional Surprise of a Base-Model

Matthew Khoriaty[1] Claude (Anthropic)[2]

March 11, 2026

[1]ERA Cambridge [2]Anthropic

Abstract

We develop a diversity metric for language model outputs that uses a trusted base model's token-level probabilities rather than embedding-space or n-gram approaches. The metric is grounded in information theory: we measure how much each successive response from a policy remains surprising to the base model even after conditioning on all previous responses. This yields a *progressive conditional surprise curve* whose shape characterizes the learnable structure among the policy's outputs — including discrete modes, shared stylistic patterns, and topic-level regularities. We summarize this curve via excess entropy (how many bits of structure the base model can extract from the responses) and pair it with a separate coherence term (how plausible the outputs are). All quantities are computed in total bits, making the metric tokenizer-agnostic; a length-normalized variant in bits/byte is also provided for use cases where responses of different lengths should be compared on equal footing. These two quantities answer orthogonal questions and are best reported jointly; their product yields a single scalar — plausibility-weighted structural diversity — that is dimensionally consistent and does not reward noise.

1 Motivation

Consider evaluating the diversity of outputs from a language model policy π (which may be a steered, finetuned, or otherwise modified model). Existing approaches typically measure diversity via embedding-space clustering [Lai et al., 2024] or surface-level n-gram statistics. These have known limitations: embedding distances may not align with semantically meaningful differences, and n-gram metrics conflate lexical variation with genuine content diversity.

We propose an alternative grounded in information theory: use a trusted base model θ with white-box access to assess how much information responses from π share with one another. The core intuition is that if π's responses are diverse, then seeing one response should not help θ predict another. If π's responses are repetitive or constrained to a narrow set of modes, then conditioning on one response should significantly reduce θ's surprise at subsequent responses.

Why a separate base model? We deliberately avoid using π to judge its own outputs. A steered model may assign high probability to all its (narrowed) outputs, masking the loss of diversity. Using an independent θ provides an external reference frame.

Important caveat. The entire framework rests on θ's ability to perform in-context learning: when conditioned on previous responses, θ must be able to recognize patterns and reduce its

surprise at subsequent similar responses. All quantities derived below — the progressive conditional surprise curve, the excess entropy, and the coherence term — measure properties of π's outputs *as perceived by* θ. If π's outputs differ only in ways that θ cannot distinguish from context, the metric underestimates diversity. Conversely, if θ hallucinates structure that is not present, it overestimates redundancy. The metric therefore measures diversity *relative to the base model's perceptual capabilities*, which is a weaker claim than measuring the true diversity of π. As base models continue to improve at in-context learning, this gap narrows.

2 Setup and Notation

Let:

- p be a prompt,
- π be the policy under evaluation,
- $r_1, r_2, \ldots, r_n \sim \pi(\cdot \mid p)$ be n i.i.d. responses sampled from π,
- θ be a trusted base model with white-box access (we can compute token-level log-probabilities),
- $|r|_{\text{tok}}$ denote the number of tokens in response r,
- $\|r\|$ denote the number of bytes in the UTF-8 encoding of response r.

We define the **cross-entropy** (total surprise) of a response r under θ:

$$-\log_2 \theta(r \mid p) = \sum_{t=1}^{|r|_{\text{tok}}} -\log_2 \theta(r^t \mid r^{<t}, p) \tag{1}$$

where r^t is the t-th token and $r^{<t}$ the preceding tokens. Units: bits. This is the total surprise of the response — a property of the string itself, independent of θ's tokenizer (since the chain rule yields the same total regardless of how the sequence is factored).

Similarly, the **conditional cross-entropy** given previous responses $r_{<k} = (r_1, \ldots, r_{k-1})$ is $-\log_2 \theta(r_k \mid r_{<k}, p)$ (bits).

For the coherence term and diversity scores (Sections 5 and 6.3), we also use the **per-byte cross-entropy rate**:

$$h_\theta(r \mid p) = \frac{-\log_2 \theta(r \mid p)}{\|r\|} \tag{2}$$

Units: bits/byte. Normalizing by byte count rather than token count makes this rate independent of θ's tokenizer, enabling comparisons across base models with different vocabularies.

In practice, computing $\theta(r_k \mid r_{<k}, p)$ requires feeding θ a formatted context containing the prompt and all previous responses (see Section 7).

3 From Mutual Information to Progressive Conditioning

3.1 Pairwise Pointwise Mutual Information

The natural starting point is the mutual information between two responses. For specific samples r_1, r_2, the relevant quantity is the **pointwise mutual information** (PMI):

$$\text{pmi}_\theta(r_1; r_2 \mid p) = \log_2 \frac{\theta(r_1 \mid r_2, p)}{\theta(r_1 \mid p)} \tag{3}$$

Note the distinction from mutual information proper: $I_\theta(R_1; R_2 \mid p) = \mathbb{E}_{r_1, r_2 \sim \pi}[\text{pmi}_\theta(r_1; r_2 \mid p)]$. PMI for a single pair can be negative, unlike MI. With n samples, the average of $\binom{n}{2}$ pairwise PMI values estimates MI.

Asymmetry. PMI is symmetric under the true joint distribution, but the quantity above is computed via autoregressive conditioning: $\theta(r_1 \mid r_2, p)$ places r_2 before r_1 in context, and there is no guarantee that swapping the order yields the same value. The asymmetry $\mathrm{pmi}_\theta(r_1; r_2 \mid p) \neq \mathrm{pmi}_\theta(r_2; r_1 \mid p)$ is an artifact of θ's context-order sensitivity. One could symmetrize by averaging both orderings, at the cost of doubling the computation. Since the progressive conditional surprise framework (Section 3.3) commits to a single ordering anyway and does not rely on pairwise PMI, this asymmetry does not affect our final metric, but it illustrates why pairwise approaches are less natural than sequential conditioning for autoregressive models.

3.2 The Coherence Problem

Low MI (or low average PMI) means responses are approximately statistically independent under θ's assessment. But there are two distinct ways to achieve independence:

1. **Diverse-coherent**: π has broad support over plausible completions, each individually likely under θ.
2. **Random/incoherent**: π produces noise; each sample is improbable under θ but trivially independent.

Both yield pmi ≈ 0. The distinguishing signal lies in the marginal $h_\theta(r \mid p)$, not in the pairwise structure. This motivates decomposing the metric into separate coherence and independence components.

3.3 Progressive Conditional Surprise

Rather than all-pairs PMI (which costs $O(n^2)$ forward passes), we use the chain rule of mutual information. Define:

$$a_k = -\log_2 \theta(r_k \mid r_{<k}, p) \quad \text{for } k = 1, \ldots, n \tag{4}$$

Units: bits. Each a_k is the total surprise of the k-th response given all previous responses. The sequence $(a_1, a_2, \ldots, a_n)$ is the **progressive conditional surprise curve**.

Since responses have different lengths, the raw a_k values for a single ordering reflect both structural learning and length variation. Averaging over random permutations of the response ordering (Section 7.3) removes the length effect: in expectation over permutations, each position averages over all responses, so the curve reflects only how θ's predictions improve with more context.

If θ were the true joint distribution of π's outputs, then $\mathbb{E}[a_k]$ would be non-increasing in k by the information-theoretic identity $H(X \mid Y, Z) \leq H(X \mid Y)$. Since a_k is a *cross*-entropy under θ rather than the true conditional entropy, this monotonicity is an empirical property of θ's in-context learning, not a theorem: the cross-entropy decomposes as $H + D_{\mathrm{KL}}$, and while the entropy component is non-increasing with additional conditioning, the model-mismatch component could in principle increase if θ's predictions degrade with longer or more complex contexts. We assume throughout that θ is a sufficiently capable model that this mismatch remains small and does not reverse the monotonicity. If a_k is observed to be non-monotone in practice, this is a diagnostic signal that θ's in-context learning is failing for the given context length or formatting. The total correlation decomposes as:

$$\mathrm{TC}_n = \sum_{k=1}^{n} \left[-\log_2 \theta(r_k \mid p) - a_k \right] \tag{5}$$

The a_k curve is the fundamental object of interest. Its shape reveals the structure among the policy's outputs:

- For small k, each new response is likely to contain patterns θ has not yet seen, so a_k remains near a_1.
- As k grows, θ begins to recognize recurring structure — shared modes, stylistic regularities, topic patterns — and a_k declines.
- For large k, a_k converges to a floor $a_\infty > 0$ reflecting irreducible within-pattern variation that θ cannot predict from context alone.

This is analogous to a scree plot in PCA: the location of the "elbow" reveals when θ has learned most of the available structure, the floor corresponds to irreducible noise variance, and the initial height reflects the total variance. Overlaying a_k curves from different policies (e.g., base vs. steered) on the same axes provides the same kind of immediate visual diagnostic that comparing scree plots does.

Curve shape depends on m/n. The qualitative shape of the a_k curve depends on the ratio of the number of modes m to the number of samples n. When m is small relative to n (e.g., 2–3 modes among 15 responses), θ discovers all modes early: even the first few responses are likely to cover most modes, so a_k begins declining immediately and the curve is *concave-up* throughout — resembling exponential decay with no identifiable elbow. When m is comparable to n (e.g., 10 modes among 15 responses), the first several responses each reveal a genuinely new mode, and θ cannot begin learning until it has accumulated enough examples. The curve is then *sigmoidal*: an initial plateau where $a_k \approx a_1$ (most modes remain unseen), followed by a rapid decline once θ has observed enough modes to begin generalizing, and finally convergence to a_∞. The inflection point of the sigmoid occurs at $k_0 \propto m$ (specifically $k_0 \approx m \ln 2$; see Appendix A for a derivation) and marks the transition from "still discovering modes" to "consolidating learned structure." The curve shape is therefore itself diagnostic: a concave-up curve signals few modes relative to the sample size, while a sigmoidal curve with a visible plateau signals many modes. This distinction matters for mode-count estimation (Section 6.5) and for the choice of estimator for a_∞ (Section 4.1).

4 Excess Entropy

4.1 The Excess Entropy

Neither the running average $\bar{a}_n = \frac{1}{n}\sum_k a_k$ nor the total correlation TC_n is a satisfactory single scalar. The running average converges to a_∞ (the uninteresting residual floor), losing sensitivity to the learnable structure above it. The total correlation grows linearly in n and never saturates, because each new term contributes approximately $-\log_2 \theta(r \mid p) - a_\infty > 0$ for large k.

The solution is to isolate the signal above the noise floor. Following the concept of excess entropy from computational mechanics [Crutchfield and Feldman, 2003], define the **excess entropy**:[1]

$$E = \sum_{k=1}^{\infty} \big(a_k - a_\infty\big) \tag{6}$$

[1]The excess entropy $E = \sum_k (a_k - a_\infty)$ is defined as a discrete sum over integer response indices, since "conditioning on 3.5 responses" has no direct meaning. However, a continuous extension exists: consider conditioning on a random subset of size αn for $\alpha \in [0, 1]$, yielding a continuous curve $\bar{a}(\alpha)$. The discrete $\bar{a}_k$ would be samples at $\alpha = k/n$, and E would correspond to integrating this curve (in which case trapezoidal integration would be more faithful than Riemann sums). We treat E as the discrete sum throughout, but note that the continuous interpretation remains an open question, particularly in connection to the continuous excess entropy literature. The practical difference between Riemann and trapezoidal sums over ∼15 points is small.

Units: bits. This is the total learnable structural information in the progressive conditioning process, above the irreducible residual noise. It converges whenever θ's surprise at new responses eventually stabilizes (since $a_k \to a_\infty$ and the excess $e_k = a_k - a_\infty$ decays to zero). Note that the "structure" captured by E is not limited to discrete mode identity: it includes any inter-response regularity that θ can learn from context, such as shared stylistic patterns, vocabulary, topic constraints, and formatting conventions, in addition to discrete modes.

In practice, a_∞ is unknown. We estimate it as a_n (the last observed value) and compute:

$$\hat{E}_n = \sum_{k=1}^{n} (a_k - a_n) \tag{7}$$

This underestimates E (since $a_n \geq a_\infty$), but the bias decreases with n.

Parametric extrapolation of a_∞. Rather than using a_n directly, one can fit a parametric model to the observed curve and extrapolate a_∞. As discussed in Section 3.3, the a_k curve is theoretically expected to be sigmoidal (with concave-up/exponential decay as the degenerate case when the inflection point $k_0 < 1$). This motivates fitting a four-parameter sigmoid:

$$a_k = a_\infty + \frac{\alpha}{1 + e^{\beta(k-k_0)}} \tag{8}$$

where a_∞ is the asymptotic floor, α is the total drop from a_1 to a_∞, $\beta > 0$ controls the steepness of the transition, and k_0 is the inflection point. This fit has several advantages: (i) it yields a_∞ without requiring n to be large enough for convergence, reducing the tension between wanting large n for a good estimate and keeping n small to avoid out-of-distribution effects; (ii) the inflection point k_0 provides a principled mode-count estimate when $k_0 > 1$ (see Section 6.5); and (iii) it unifies the exponential and sigmoidal regimes — when the fit returns $k_0 < 1$, the curve is effectively exponential (few modes), and when $k_0 > 1$, the curve has a genuine plateau (many modes). In practice, we recommend fitting Eq. 8 by nonlinear least squares and using the fitted a_∞ in place of a_n in Eq. 7, particularly when the curve has not visibly converged by $k = n$.

Out-of-distribution risk. In practice, the conditioning context grows with k. For large k, the concatenation of many responses may push θ out of distribution: the context may exceed lengths seen during pretraining, or the repeated-response formatting may be sufficiently unusual that θ's predictions become unreliable. When θ goes out of distribution, a_k may *increase* rather than decrease (reflecting model confusion rather than genuine novelty), violating the monotonicity assumption and corrupting the excess entropy estimate. It is therefore important to monitor a_k for non-monotonicity, particularly at large k, and to keep n within a range where the total context length remains well within θ's reliable operating regime. If responses are long, n may need to be reduced accordingly.

Relationship to total correlation. The excess entropy and total correlation are complementary decompositions of the same underlying structure. Working in expectation over i.i.d. draws from π, let $\bar{a}_1 = \mathbb{E}[-\log_2 \theta(r \mid p)]$ denote the expected unconditional cross-entropy in bits (identical for all k by identical distribution), $\bar{a}_k = \mathbb{E}[a_k]$ the expected conditional cross-entropy at position k, and $e_k = \bar{a}_k - a_\infty$ the excess at step k.

The per-step mutual information — how much the previous responses tell θ about response k — is:

$$I_k = \bar{a}_1 - \bar{a}_k \tag{9}$$

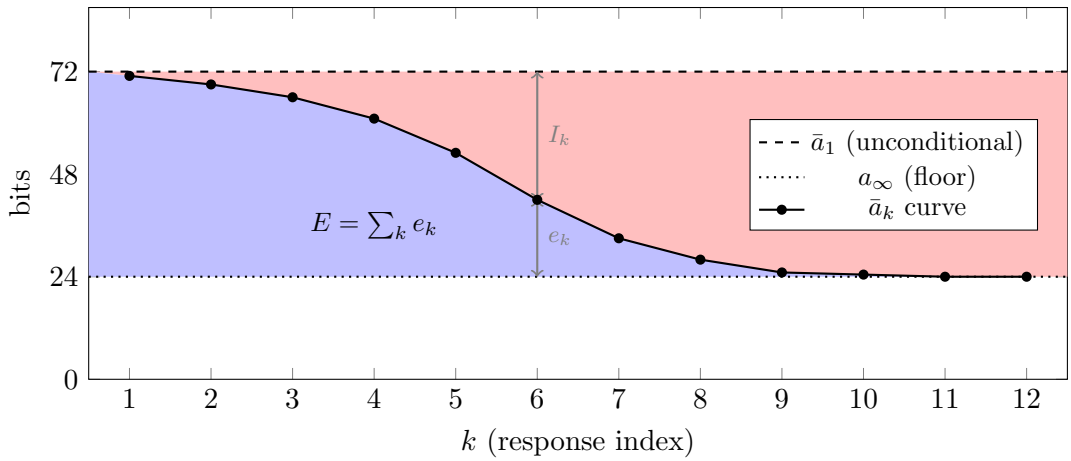

Figure 1: Decomposition of the gap between unconditional surprise $\bar{a}_1$ and the asymptotic floor a_∞ at each step k. The per-step gap $\bar{a}_1 - a_\infty$ splits into mutual information $I_k = \bar{a}_1 - \bar{a}_k$ (red, above the curve) and excess $e_k = \bar{a}_k - a_\infty$ (blue, below the curve). Summing across k: the red area is the total correlation TC_n; the blue area is the excess entropy E. As k grows, $e_k \to 0$ and each step contributes the full $\bar{a}_1 - a_\infty$ to TC_n, so TC_n grows linearly while E converges. All quantities are in bits.

Substituting $\bar{a}_k = a_\infty + e_k$:

$$I_k = (\bar{a}_1 - a_\infty) - e_k \tag{10}$$

The total correlation is the sum of per-step MI:

$$\mathrm{TC}_n = \sum_{k=1}^{n} I_k = n(\bar{a}_1 - a_\infty) - \sum_{k=1}^{n} e_k = n(\bar{a}_1 - a_\infty) - E_n \tag{11}$$

Equivalently:

$$E_n = n(\bar{a}_1 - a_\infty) - \mathrm{TC}_n \tag{12}$$

For large n where $E_n \to E$, the total correlation grows linearly with slope $(\bar{a}_1 - a_\infty)$:

$$\mathrm{TC}_n \approx n(\bar{a}_1 - a_\infty) - E \tag{13}$$

The three quantities have clean interpretations, all in bits: $(\bar{a}_1 - a_\infty)$ is the per-response redundancy once θ has fully learned the available structure (the asymptotic MI rate); E is the total finite structural information, consumed during θ's transient learning phase; and TC_n is the cumulative redundancy, which grows without bound because even after all learnable structure has been extracted, each new response is partially predictable.

Figure 1 illustrates this decomposition. At each step k, the gap between a_∞ and $\bar{a}_1$ decomposes into I_k (how much θ has learned by step k) and e_k (how much remains to be learned). The excess entropy E is the total area of the blue region; the total correlation TC_n is the total area of the red region.

4.2 Per-Byte Excess Entropy Rate

The excess entropy $E = \sum_k (a_k - a_\infty)$ is in bits and serves as the primary scalar measure of learnable structure. For the length-normalized diversity score D (Section 6.3), we define the **per-byte excess**

entropy rate by normalizing each response's surprise by its byte count *before* averaging across permutations.

For a single permutation σ, the per-byte conditional surprise at position k is $a_k^\sigma / \|r_{\sigma(k)}\|$ (bits/byte). Averaging over permutations gives a permutation-averaged per-byte curve:

$$\hat{a}_k^{\text{rate}} = \mathbb{E}_\sigma \left[\frac{a_k^\sigma}{\|r_{\sigma(k)}\|} \right] \tag{14}$$

The per-byte floor is estimated analogously from the last position: $\hat{a}_\infty^{\text{rate}} = \hat{a}_n^{\text{rate}}$. The per-byte excess entropy rate is then:

$$E_{\text{rate}} = \sum_{k=1}^{n} \left(\hat{a}_k^{\text{rate}} - \hat{a}_\infty^{\text{rate}} \right) \tag{15}$$

Units: bits/byte. This treats each response equally regardless of length: a 50-byte and a 500-byte response at the same per-byte excess rate contribute equally to E_{rate}, which is the correct behavior for a length-normalized score.

Note that $E_{\text{rate}} \neq E/\bar{B}$ in general, because $\mathbb{E}[X/Y] \neq \mathbb{E}[X]/\mathbb{E}[Y]$ when X and Y are correlated. Total surprise a_k and byte count $\|r_k\|$ are positively correlated (longer responses have more tokens to predict), so $E/\bar{B}$ overweights long responses. The per-response normalization in E_{rate} avoids this.

5 The Coherence Term

5.1 Does Excess Entropy Already Handle Incoherence?

At first glance, excess entropy appears to handle the coherence problem from Section 3.2 on its own. A policy that outputs pure noise has $a_k \approx \text{const}$ for all k (random tokens from one response do not help predict random tokens from another), so $e_k \approx 0$ and $E \approx 0$. The noise policy correctly scores as having zero structural diversity. No separate coherence penalty is needed for this case.

The problem arises with a **policy that has multiple distinct but individually incoherent modes**. Consider a badly steered or finetuned model that produces several recognizably different types of low-quality output. The base model θ can learn from context which type is being produced (so a_k drops meaningfully across k), but each type is still implausible text. In this case $E > 0$, yet the policy should not score as meaningfully diverse. This failure mode is particularly relevant when evaluating models at high steering magnitudes, where the model may produce incoherent outputs that nonetheless condition strongly on their own prior responses.

5.2 Definition

The per-byte cross-entropy $h_\theta(r \mid p)$ from Eq. (2) can be converted to a geometric mean per-byte probability:

$$2^{-h_\theta(r|p)} = \left(\prod_{t=1}^{|r|_{\text{tok}}} \theta(r^t \mid r^{<t}, p) \right)^{1/\|r\|} \tag{16}$$

This is the average per-byte probability the base model assigns to the response, living in $(0, 1)$.

We define **coherence** as the geometric mean of this quantity across responses:

$$C = 2^{-\frac{1}{n} \sum_{i=1}^{n} h_\theta(r_i|p)} \tag{17}$$

This is interpretable as "how likely, per byte, does the base model find this policy's outputs?" Equivalently, C is the reciprocal of the geometric-mean per-byte perplexity:

$$C = \frac{1}{\mathrm{PPL}_\theta(\pi, p)}, \qquad \mathrm{PPL}_\theta(\pi, p) = 2^{\frac{1}{n}\sum_{i=1}^{n} h_\theta(r_i|p)} \tag{18}$$

We chose the geometric mean (equivalently, $1/n$ inside the exponent) over the arithmetic mean $C_{\mathrm{AM}} = \frac{1}{n}\sum_i 2^{-h_\theta(r_i|p)}$. By Jensen's inequality (since 2^{-x} is convex), $C_{\mathrm{AM}} \geq C$ with equality iff all $h_\theta(r_i \mid p)$ are identical. The gap is precisely what makes the arithmetic mean problematic: if 9 out of 10 responses are incoherent and one is highly plausible, C_{AM} remains nonzero while C is driven toward zero by the geometric averaging.

Distributional distance. In expectation over $r \sim \pi$:

$$\mathbb{E}_\pi\big[h_\theta(r \mid p)\big] = H_\pi(\cdot \mid p) + D_{\mathrm{KL}}(\pi \| \theta \mid p) \tag{19}$$

So coherence is penalized both by genuine incoherence and by distributional distance from the base model. For the purpose of evaluating steered or finetuned policies, this is a feature: a heavily steered policy that assigns probability mass to regions θ finds implausible is penalized accordingly, reflecting that the base model's probability landscape is treated as the reference.

Note on comparability. Because all quantities are normalized per byte, the per-byte coherence C is invariant to tokenizer choice: two base models θ_1, θ_2 with different tokenizers but identical predictive distributions will yield identical C values. What remains incomparable across different θ's is model quality: a better model assigns lower cross-entropy regardless of normalization. This is by design — θ's predictive quality is the lens through which diversity is measured.

5.3 Why Not Weight Excess Entropy by Coherence?

A natural alternative to reporting E and C separately is to weight the terms within E by coherence, computing $E_w = \sum_k w_k \cdot (a_k - a_\infty)$ where $w_k = 2^{-h_\theta(r_k|p)}$ is the per-response coherence. This would suppress contributions from incoherent modes directly at the point of measurement. We considered and rejected this approach for three reasons.

First, it breaks the information-theoretic interpretation. $E = \sum_k e_k$ has a clean meaning as the total structural information above the noise floor, connected to the excess entropy of Crutchfield and Feldman [2003]. Inserting weights makes E_w a hybrid quantity that is not the excess entropy of any well-defined process.

Second, the weights are not independent of what they weight. The weight w_k uses $h_\theta(r_k \mid p)$ and the excess e_k uses $-\log_2 \theta(r_k \mid r_{<k}, p)$. Both are derived from the same response r_k scored by the same model θ. An incoherent response from a novel mode has both high e_k (novel) and low w_k (incoherent), which partially cancel, but *how much* they cancel depends on the specific relationship between unconditional and conditional surprise for that response, which varies in ways that lack a clean characterization.

Third, it makes the noise floor a_∞ ambiguous. In the unweighted version, a_∞ is simply the asymptotic conditional surprise. In the weighted version, it is unclear whether a_∞ should also be weighted, and if so, by what. The separate-terms approach avoids this because a_∞ lives purely in the information-theoretic layer and C lives purely in the plausibility layer.

6 Reporting the Metric

6.1 The Curve Is the Metric

The a_k curve is the primary output of this framework. It reveals the total learnable structure among the policy's outputs: the elbow location indicates when θ has learned most of the available patterns, a_∞ reflects residual unpredictable variation, and a_1 reflects the unconditional surprise. Overlaying curves from different policies on the same axes provides immediate visual diagnosis of how an intervention (steering, finetuning) changes the output distribution, just as comparing scree plots does in PCA.

Everything below is a lossy summary of this curve. We develop these summaries because curves are difficult to rank, average across prompt suites, or report in tables, but the reader should understand that each successive compression discards information.

6.2 The (E, C, σ_ℓ) Triple

The excess entropy E and the per-byte coherence C answer genuinely different questions:

- E (bits): "How many bits of learnable structure does θ find among π's outputs?"
- C (dimensionless, in $(0, 1)$): "How likely, per byte, does θ find π's outputs?"

Since (E, C) is a two-dimensional summary, any scalar function $f(E, C)$ discards information: knowing both E and C always permits recovering the scalar, but not vice versa.

This separation enables diagnostic reasoning. For instance, when comparing a base policy against a steered variant, one might observe that steering reduces E (less learnable structure) while maintaining C (individually coherent outputs), or at high steering magnitudes, that C drops (incoherent outputs) while E remains nonzero (multiple types of incoherent output). These are qualitatively different failure modes that a single scalar would conflate.

Because C is a mean, it can mask heterogeneity. We therefore pair it with the coherence spread σ_ℓ (Section 6.4), giving the standard summary triple (E, C, σ_ℓ). Note that σ_ℓ is defined in per-byte units (the standard deviation of per-byte cross-entropies $h_\theta(r_i \mid p)$) rather than total bits, because this isolates quality variation from length variation, making it a better diagnostic of coherence heterogeneity.

6.3 Scalar Summary: Plausibility-Weighted Structural Diversity

When a single number is required (e.g., for ranking policies in a table), we combine coherence and excess entropy into a diversity score. This can be done at two granularities.

The **diversity score**:

$$D = C \times E \tag{20}$$

Units: bits. This is the per-byte-coherence-weighted structural information across all responses: "how many bits of learnable structure does θ find, weighted by how plausible per byte the outputs are?" Because E is not length-normalized, longer responses can score higher even at the same diversity rate, which is appropriate when longer responses genuinely carry more diverse content.

The **per-byte diversity score**:

$$D_{\text{rate}} = C \times E_{\text{rate}} \tag{21}$$

Units: bits/byte. This is coherence-weighted structural information per byte: "how much learnable structure is there per byte, discounted by how plausible the outputs are?" Because both C and E_{rate} are length-normalized, D_{rate} treats long and short responses on equal footing.

The two scores are complementary. D is the right choice when the total amount of diverse information matters regardless of response length. D_{rate} is the right choice when comparing policies that may produce responses of different lengths on the same prompt and length variation should be normalized away.

Typical range of C. For well-formed English text scored by a capable base model, bits-per-byte values typically fall in the range 0.5–1.5 (corresponding to per-byte perplexities of roughly 1.4–2.8), so $C \in [0.35, 0.7]$. This is considerably less compressed than the per-token range, because bytes are a finer-grained unit than tokens. Consequently, $D = C \times E$ is not dominated by C in the way a total-probability coherence term would be. Nonetheless, two mitigation strategies remain available for cases where coherence variation dominates diversity variation:

1. **Normalization**: Replace C with C/C_{base}, where C_{base} is the coherence of θ's own samples given the same prompt. The ratio lives in $(0, 1]$ and equals 1 when π is as coherent as θ, removing the baseline suppression. This has a clean interpretation as "what fraction of baseline coherence does π retain?"
2. **Fractional exponent**: Replace C with C^α for $\alpha \in (0, 1)$, which compresses the coherence penalty. This softens the dominance of C but introduces a free hyperparameter without information-theoretic motivation.

We recommend normalization when a single-scalar ranking is the primary use case, and the unmodified C when the goal is diagnostic (reporting E and C separately).

The edge cases behave correctly for both D and D_{rate}:

1. **Pure noise**: $C \approx 0$, $E \approx 0$, so $D \approx 0$ and $D_{\text{rate}} \approx 0$.
2. **Multiple incoherent patterns**: C low, $E > 0$, so both scores are suppressed.
3. **Rich coherent structure**: C high, E high, so both scores are high.
4. **One coherent pattern**: C high, $E \approx 0$, so both scores are ≈ 0.
5. **Decomposable**: One can always factor D back into C and E (or D_{rate} into C and E_{rate}) to identify which term drove a change.

Limitations of the scalar summary. D inherits two limitations from E. First, E measures learnable *structure* — recurring patterns that θ can extract from seeing multiple responses — not continuous spread. A policy that draws from a broad, roughly continuous distribution of coherent outputs has $a_k \approx a_1$ for all k, because no recurring pattern exists for θ to learn. This yields $E \approx 0$ and $D \approx 0$ despite maximal diversity. In practice, LLM output distributions tend toward structured output (stories cluster into genres, arguments cluster into strategies, and shared stylistic conventions recur), so this is unlikely to cause problems for typical evaluations. The diagnostic is high C with low E: if π's outputs are individually plausible but θ finds no inter-response structure, the diversity may be continuous rather than absent. The a_k curve distinguishes these cases by its height (a_1), which D discards.

Second, E measures statistical independence of strings as perceived by θ, not semantic diversity of content. A policy that always conveys the same information in different phrasings would ideally score as low-diversity. We expect a capable θ to compress paraphrases effectively — recognizing "I have seen this fact/plot before" from context is precisely the kind of in-context learning the metric relies on — but the degree of compression is an empirical property of θ. This is arguably a strength over embedding-based diversity metrics, which may register paraphrases as distant because the surface strings differ.

6.4 Coherence Heterogeneity

The coherence C summarizes the plausibility of π's outputs as a single number, but a policy may mix coherent and incoherent modes. A policy with 5 coherent modes and 3 incoherent modes will have E reflecting all 8 modes (since θ learns to predict all of them from context), while C is dragged down by the incoherent samples. The pair (E, C) cannot distinguish "8 modes, 3 of which are junk" from "8 uniformly mediocre modes."

To diagnose this, we report the **coherence spread** σ_ℓ, defined as the standard deviation of the per-response cross-entropies:

$$\sigma_\ell = \mathrm{std}\big(\{h_\theta(r_i \mid p)\}_{i=1}^n\big) \tag{22}$$

Since $C = 2^{-\bar{\ell}}$ where $\bar{\ell} = \frac{1}{n}\sum_i h_\theta(r_i \mid p)$, the spread σ_ℓ induces a multiplicative coherence interval. Define:

$$C_+ = 2^{-(\bar{\ell}-\sigma_\ell)} = C \cdot 2^{\sigma_\ell}, \qquad C_- = 2^{-(\bar{\ell}+\sigma_\ell)} = C \cdot 2^{-\sigma_\ell} \tag{23}$$

C_+ is the coherence of the better-than-average responses and C_- is the coherence of the worse-than-average responses. These yield a **diversity uncertainty band**:

$$D_\pm = C_\pm \times E_{\mathrm{rate}} \tag{24}$$

The width of this band, $D_+ - D_- = C(2^{\sigma_\ell} - 2^{-\sigma_\ell}) \times E_{\mathrm{rate}}$, is zero when all responses have equal coherence and large when the policy mixes coherent and incoherent modes. A wide band is the diagnostic signal for heterogeneous output quality.

What to report. We recommend reporting (E, C, σ_ℓ) as the standard summary triple. When σ_ℓ is small, C faithfully represents the policy's coherence and $D = C \times E$ is a reliable scalar. When σ_ℓ is large, the band $[D_-, D_+]$ conveys the uncertainty, and the policy merits closer inspection (e.g., examining which modes are incoherent via the per-response $c_i = 2^{-h_\theta(r_i|p)}$ values).[2]

6.5 Estimating Mode Count from the Curve

While E measures the total structural information (in bits) that θ can learn from π's outputs, it does not directly yield a mode count. The naïve formula $m_{\mathrm{eff}} = 2^E$ assumes that all learnable structure corresponds to mode identity, which holds only for near-deterministic modes. For real LLM outputs, θ learns far more from context than which mode is active — it learns shared stylistic patterns, vocabulary, topic-specific structure, and other within-mode regularities. This inflates E well beyond $\log_2 m$, making m_{eff} unreliable.

The number of effective modes is instead read from the **elbow of the a_k curve**. While θ is still discovering new modes ($k \lesssim m$), each response is likely novel and a_k remains near a_1. Once θ has seen examples from most modes ($k \gtrsim m$), a_k drops toward a_∞. The transition — the elbow — occurs at $k \approx m$, regardless of how much within-mode structure exists. This can be identified visually when comparing a small number of policies, or via changepoint detection methods when automated estimation is needed.

[2] Two more aggressive diagnostic strategies are available when σ_ℓ is large. First, *threshold-and-recompute*: set a coherence floor $c_{\min}$ (e.g., derived from θ's own output distribution), discard responses with $c_i < c_{\min}$, and recompute E on the surviving subset. This directly answers "how diverse is π among its coherent outputs?" at the cost of introducing a hyperparameter and reducing the effective sample size. Second, *coherence-stratified a_k curves*: sort responses by coherence, then compute separate a_k curves for the top-k most coherent responses and for the full set. If the curves diverge, incoherent modes are inflating E. Both strategies are complementary to the (E, C, σ_ℓ) triple and can be applied post hoc without modifying the core metric.

Caveat: elbow detection requires m comparable to n. The elbow prescription assumes that the a_k curve has a visible plateau followed by a decline. As discussed in Section 3.3, this sigmoidal shape only appears when m is comparable to n. When $m \ll n$ (e.g., 2–3 modes among 15 samples), θ discovers all modes within the first few responses, and the curve is monotonically concave-up (exponential decay) with no identifiable elbow or plateau. In this regime, mode count cannot be directly read from curve shape. The parametric sigmoid fit (Eq. 8) provides an alternative: the fitted inflection point k_0 estimates m even when no visual elbow is present (returning $k_0 < 1$ for the few-mode case, correctly indicating that mode discovery is essentially immediate).

6.6 Aggregation Across Prompts

The diversity scores are defined relative to a single prompt p. To summarize a policy's diversity across a prompt suite $P = \{p_1, \ldots, p_M\}$, we average. For the diversity score:

$$\bar{D}(\pi) = \frac{1}{M} \sum_{j=1}^{M} D(\pi, p_j) \tag{25}$$

To compare two policies (e.g., a base model against a steered variant), the natural scalar is the difference:

$$\Delta D = \bar{D}(\pi_{\text{base}}) - \bar{D}(\pi_{\text{steered}}) = \frac{1}{M} \sum_{j=1}^{M} \big[D(\pi_{\text{base}}, p_j) - D(\pi_{\text{steered}}, p_j) \big] \tag{26}$$

ΔD answers "how many bits of structural diversity did steering remove, on average?" For constrained prompts where both policies have $D \approx 0$, the per-prompt contribution is ≈ 0, which is correct: there was no diversity to remove. For open-ended prompts where steering collapses structure, the contribution is large and positive. No division is involved, so ΔD is stable even when D_{base} is near zero.

Length-normalized aggregation. The per-byte score D_{rate} can be aggregated analogously: $\bar{D}_{\text{rate}} = \frac{1}{M} \sum_j D_{\text{rate}}(\pi, p_j)$ and $\Delta D_{\text{rate}} = \bar{D}_{\text{rate,base}} - \bar{D}_{\text{rate,steered}}$, with ΔD_{rate} in bits/byte. The choice between D and D_{rate} for aggregation depends on whether length variation across prompts should be counted as part of the diversity signal (use D) or normalized away so that long and short responses are compared on equal footing (use D_{rate}).

Why not a ratio? The relative measure $D_{\text{steered}}/D_{\text{base}}$ is appealing ("what fraction of diversity survived steering?") but requires division by D_{base}, which is near zero for any constrained prompt. Clamping the denominator (e.g., $\max(D_{\text{base}}, \epsilon)$) introduces a hyperparameter and makes the ratio depend on ϵ precisely for the prompts where it matters most. The absolute difference avoids these issues.

6.7 What to Report

In order of informativeness:

1. The a_k curve (the primary metric), which reveals the total learnable structure, its rate of acquisition, residual variation (a_∞), and unconditional surprise (a_1). The elbow location indicates approximate mode count (Section 6.5). When comparing policies, overlaying curves on the same axes provides immediate visual insight.

2. The (E, C, σ_ℓ) triple, for tabular comparisons across many policies or prompts. When σ_ℓ is large, report the diversity band $[D_-, D_+]$.
3. $D = C \times E$ (bits) when a single per-prompt number is needed and σ_ℓ is small. Alternatively, $D_{\text{rate}} = C \times E_{\text{rate}}$ (bits/byte) when long and short responses should be compared on equal footing.
4. $\Delta D = \bar{D}_{\text{base}} - \bar{D}_{\text{steered}}$ (or ΔD_{rate}) when comparing two policies across a prompt suite.

7 Practical Considerations

7.1 Formatting the Conditioning Context

Computing $\theta(r_k \mid r_{<k}, p)$ requires feeding θ a context containing the prompt and previous responses. The formatting choice affects results. We recommend using a base (non-instruction-tuned) model as θ and formatting as:

```
[prompt p]

Response A: [r_1]

Response B: [r_2]

Response C: [r_3]
```

This leverages the base model's in-context learning to treat the responses as parallel completions. An instruction-tuned model might instead interpret previous responses as conversation turns, confounding the measurement. The formatting choice should be ablated.

Base vs. instruction-tuned models. There is a tension between two desiderata for θ. On one hand, an independent base model provides a cleaner external reference frame: instruction-tuned models have had their output distribution shaped by RLHF or similar procedures, so using one as θ reintroduces the kind of distributional bias we seek to avoid. On the other hand, instruction-tuned models are specifically trained for structured multi-example reasoning, which could make them better at the in-context pattern recognition the metric requires.

We believe base models are the better default for three reasons. First, modern base models already exhibit strong in-context learning — the capability emerges from pretraining and does not require instruction tuning. Second, the formatting of the conditioning context (labeling responses as "Response A", "Response B", etc.) can be designed to elicit ICL behavior from base models via few-shot-style prompting, effectively leveraging their ICL capabilities without the distributional confounds of instruction tuning. Third, an instruction-tuned θ may assign systematically different probabilities to text that follows or violates its alignment training, introducing a confound between coherence-as-fluency and coherence-as-alignment that the metric should not conflate.

That said, if the base model's ICL proves inadequate for a particular evaluation (diagnosed by non-monotone a_k curves), an instruction-tuned θ is a reasonable fallback. The key requirement is consistency: the same θ must be used across all policies being compared.

7.1.1 Dimension-Specific Prompt Structures

The neutral formatting above ("Response A:", "Response B:", ...) lets θ attend to whatever inter-response structure it finds most salient. An alternative is to use *dimension-specific* framing that

primes θ to attend to a particular axis of variation. For example:

- "Here are several stories in different genres:" $\rightarrow \theta$ attends to genre structure.
- "Here are several responses with different writing styles:" $\rightarrow \theta$ attends to stylistic variation.
- "Here are several arguments for different positions:" $\rightarrow \theta$ attends to stance diversity.

The resulting E then measures diversity *along the specified dimension as perceived by* θ. By computing E under multiple framings, one obtains a **diversity profile**: a vector of per-dimension diversity scores for the same set of responses. This could reveal, for instance, that a happiness steering vector collapses topic diversity but preserves stylistic diversity, a richer claim than "steering reduces diversity" in aggregate.

We recommend the neutral prompt structure as the default for overall diversity measurement. Dimension-specific structures are a complementary tool for targeted analysis. Three caveats apply. First, each measurement is now $(\theta, \text{framing})$-relative rather than just θ-relative, adding a degree of freedom that must be controlled across comparisons. Second, a leading framing can cause θ to hallucinate structure that is not present — an aggressive prompt like "Here are responses with hidden symbolic patterns:" might inflate E for random text. The neutral framing avoids this by not priming θ toward any particular kind of structure, and should always be reported alongside any dimension-specific measurements. Third, the effect of the framing is concentrated in the early a_k values. As θ sees more responses, the actual structure in the data dominates the prior imposed by the framing, and a_k curves from different prompt structures should converge. This means that dimension-specific framings primarily affect *what* θ *learns first*, not the asymptotic floor or the total excess entropy at large n. The meaningful differences between diversity profiles are therefore found in the first few values of the a_k curve, and using large n may wash out the signal that dimension-specific framing is intended to capture.

7.2 Computational Cost

Because θ is a causal language model, the full a_k curve can be computed from a **single forward pass**. Concatenate the prompt and all responses into one sequence:

```
[prompt p] Response A: [r_1] Response B: [r_2] ... Response N: [r_n]
```

The forward pass produces the log-probability of every token conditioned on all preceding tokens. The tokens belonging to r_k are conditioned on exactly $p, r_1, \ldots, r_{k-1}$ (plus formatting), which is the conditioning required for a_k. Partitioning the output log-probabilities by response boundary and summing within each partition yields all n values of a_k simultaneously. The cost is a single forward pass over the full concatenation, with FLOPs scaling as $O((|p| + n\bar{L}_{\text{tok}})^2)$ from causal attention, where $\bar{L}_{\text{tok}}$ is the average response length in tokens.

The coherence term C and spread σ_ℓ additionally require the *unconditional* per-byte cross-entropies $h_\theta(r_i \mid p)$ for each response. These cannot be extracted from the concatenated pass, since in that pass r_i for $i > 1$ is conditioned on all prior responses. Computing them requires n independent forward passes, each over the short context (p, r_i). These passes are embarrassingly parallel and batchable.

In summary:

- a_k curve and E: 1 forward pass (long context).
- C and σ_ℓ: n forward passes (short contexts, batchable).
- Permutation averaging (if desired): multiply the single-pass cost by the number of permutations.

This compares favorably to all-pairs PMI, which requires $O(n^2)$ forward passes.

7.3 Dependence on Sample Ordering

The a_k values depend on the ordering of $\{r_i\}$. Since a_k is in total bits, responses of different lengths produce different a_k values even at the same per-byte rate, making a single ordering's curve jagged. Averaging over random permutations removes this length effect: each position averages over all responses, so the curve reflects only how θ's predictions improve with more context. In expectation over random permutations the summary statistics (E, D) are invariant. We recommend averaging over a small number of random permutations (e.g., 3–5), which substantially reduces variance at a proportional increase in cost.

For the per-byte excess entropy rate E_{rate} (Section 4.2), per-response normalization by byte count must be performed *before* averaging across permutations, since the byte count depends on which response lands at each position.

7.4 Choice of n

The excess entropy estimator $\hat{E}_n$ depends on n through both the summation range and the estimate $a_n \approx a_\infty$. We recommend choosing n large enough that a_n has approximately stabilized (i.e., the curve has reached its floor), and fixing n across all comparisons for consistency. A practical choice is $n = 10$–20, which suffices for policies where θ's learning stabilizes within this range.

Detectable mode weight. Modes with weight $w \lesssim 1/n$ will rarely appear in the sample, and even when they do, θ has insufficient examples to learn the pattern. As a rough guide, modes with weight below $1/n$ are invisible to the metric. For $n = 15$, this means modes contributing less than $\sim$7% of the probability mass are unlikely to be captured. If rare modes are of interest (e.g., when evaluating whether steering eliminates low-frequency behaviors), n should be increased accordingly, subject to the context-length constraints discussed above.

7.5 Ensembling Base Models

A single base model θ may produce non-monotone a_k curves when pushed out of distribution by long or unusual conditioning contexts. Ensembling multiple base models can stabilize the estimates.

The principled approach is to ensemble at the **token level**: at each position t, form the mixture distribution

$$\bar{\theta}(r^t \mid r^{<t}, \text{context}) = \frac{1}{M} \sum_{j=1}^{M} \theta_j(r^t \mid r^{<t}, \text{context}) \tag{27}$$

and compute log-probabilities under $\bar{\theta}$. This mixture is itself a valid autoregressive model. By Jensen's inequality (log is concave), $-\log_2 \bar{\theta} \leq -\frac{1}{M} \sum_j \log_2 \theta_j$: the mixture always assigns at least as much probability as the geometric mean of the individual models. If one model goes out of distribution and assigns near-zero probability to the next token, the mixture is rescued by any other model that remains well-calibrated. This is the key robustness property.

The naïve alternative — computing a_k curves independently per model and averaging them — lacks this robustness. If multiple models go out of distribution on the same context (which is likely since they share similar failure modes), their a_k values all increase simultaneously, and averaging divergent values still yields divergence. Token-level ensembling avoids this because even a single well-calibrated model in the mixture prevents the per-token probabilities from collapsing.

Tokenizer compatibility. Token-level ensembling requires all models to share a tokenizer, since the mixture averages distributions over the vocabulary. Fortunately, several cross-family groups share tokenizers: Llama 2, Mistral 7B/Large/Mixtral, Yi, and Solar all use the same SentencePiece BPE tokenizer (32K vocabulary); Gemma and Gemini share Google's SentencePiece tokenizer. Cross-family ensembles are preferable to within-family ensembles (e.g., Llama 3 8B + 70B), because models from different families have different training data and architectures, reducing the correlation of their failure modes. A practical recommendation is to ensemble a smaller and a larger model from different families that share a tokenizer: the smaller model is cheaper but more likely to fail on long contexts, while the larger model provides robustness.

Fallback for incompatible tokenizers. When the desired models use different tokenizers, one can average the total-bits a_k values across models, since total log-probability is tokenizer-agnostic. This is not a principled probabilistic mixture and does not provide the robustness guarantee of token-level ensembling, but it may still reduce variance from individual model quirks when the models are not simultaneously out of distribution.

8 Related Concepts

Excess entropy. The quantity $E = \sum_k (a_k - a_\infty)$ is related to the excess entropy of stationary stochastic processes studied in computational mechanics [Crutchfield and Feldman, 2003]. In that setting, it measures the mutual information between the past and future of a process, or equivalently the statistical complexity beyond the irreducible entropy rate. Our application differs in that the "process" is a sequence of i.i.d. responses from π, assessed by θ, rather than a single time series, and the conditioning is performed via the base model's in-context learning rather than explicit state estimation.

Perplexity. The coherence term $C = 1/\text{PPL}$ (Section 5) connects this framework to the standard language modeling evaluation metric. The diversity score $D = C \times E$ operates in bits, while the per-byte diversity score $D_{\text{rate}} = C \times E_{\text{rate}}$ provides a length-normalized variant.

Total correlation. The total correlation TC_n relates to the excess entropy via Eq. (12): $\text{TC}_n = n(\bar{a}_1 - a_\infty) - E_n$. See Section 4.1 and Figure 1 for the full derivation and a visual decomposition. Note that since all quantities are cross-entropies under θ rather than true entropies of π's output distribution, TC_n is a θ-relative quantity that measures the total correlation *as perceived by* θ; it equals the true total correlation only to the extent that θ approximates π's joint distribution.

Embedding-based diversity. Prior work uses embedding distances or clustering (e.g., k-LLMmeans [Lai et al., 2024]) to measure diversity. These approaches are complementary: they capture semantic distance in a learned representation space, while our metric captures statistical independence under a generative model. The approaches may disagree when θ's in-context learning captures structure invisible to the embedding model, or vice versa.

Sampling diversity metrics. Work on decoding-time diversity [Zhang et al., 2024] typically operates at the surface level (n-gram overlap, self-BLEU). Our metric operates at the distributional level and can distinguish policies that produce lexically varied but semantically redundant outputs.

A Relating the Sigmoid Inflection Point to Mode Count

A.1 Setup

We have m modes with uniform weights $w_i = 1/m$. Each of n responses is drawn independently from one of the m modes. We fit a sigmoid to the permutation-averaged progressive conditional surprise curve:

$$\bar{a}_k = a_\infty + \frac{\alpha}{1 + e^{\beta(k-k_0)}} \tag{28}$$

where k_0 is the inflection point, $\alpha = \bar{a}_1 - a_\infty$ is the total learnable gap, $\beta > 0$ controls the steepness of the transition, and a_∞ is the asymptotic floor. We seek the relationship between k_0 and m.

A.2 The Mode Discovery Crossover

After k responses drawn uniformly from m modes, the expected probability that the next response comes from a mode not yet observed is:

$$p_{\text{new}}(k) = \left(1 - \frac{1}{m}\right)^k \tag{29}$$

This decreases monotonically from $p_{\text{new}}(0) = 1$ toward 0. The complementary probability $1 - p_{\text{new}}(k)$ is the chance that the next response comes from an already-seen mode.

Define k^* as the **discovery crossover**: the number of responses at which a new response is equally likely to come from a novel mode or a previously seen mode. Setting $p_{\text{new}}(k^*) = 1/2$:

$$\left(1 - \frac{1}{m}\right)^{k^*} = \frac{1}{2} \tag{30}$$

$$k^* = \frac{\ln 2}{-\ln(1 - 1/m)} = \frac{\ln 2}{\ln\left(\frac{m}{m-1}\right)} \tag{31}$$

For large m, using $\ln(1 - 1/m) \approx -1/m$:

$$k^* \approx m \ln 2 \approx 0.693\, m \tag{32}$$

This is an exact combinatorial quantity. Before k^*, mode discovery dominates: most new responses reveal modes θ has not yet seen. After k^*, mode refinement dominates: most new responses come from modes θ has already encountered.

For reference, the expected number of distinct modes observed at k^* is:

$$\mathbb{E}[\text{distinct modes at } k^*] = m\left(1 - \left(1 - \frac{1}{m}\right)^{k^*}\right) = m \cdot \frac{1}{2} = \frac{m}{2} \tag{33}$$

That is, at the crossover, exactly half the modes have been seen in expectation.

A.3 Hypothesis: $k_0 \approx k^*$

We hypothesize that the sigmoid inflection point k_0 is near the discovery crossover k^*. The reasoning is as follows. The inflection point of the learning curve is where θ's marginal learning rate is greatest. Before k^*, θ is still in the discovery-dominated regime: it encounters new modes frequently, but has seen too few examples of any single mode to fully learn within-mode structure. After k^*, θ is in the

refinement-dominated regime: it rarely encounters new modes, and each additional example of a known mode provides diminishing returns. The learning rate should peak near the transition, where θ has seen enough modes to start forming generalizations and enough repeats to start exploiting them.

This is a qualitative argument, not a proof. The true inflection point depends on how much information θ extracts from a mode discovery versus a mode refinement, which is a property of θ and the mode structure. The robust prediction is:

$$k_0 \propto m \tag{34}$$

The proportionality constant is $\ln 2 \approx 0.693$ under the hypothesis $k_0 = k^*$, but the true constant is empirical. If θ is powerful enough to learn modes from very few examples, mode discovery alone drives learning and $k_0 < k^*$. If θ requires many repeated observations to characterize a mode, $k_0 > k^*$.

A.4 Mode Count Estimator

If the hypothesis holds, the fitted k_0 yields a mode count estimator:

$$\hat{m} = \frac{k_0}{\ln 2} \approx 1.443\, k_0 \tag{35}$$

This assumes uniform mode weights. When $k_0 < 1$ (the sigmoid degenerates to exponential decay), the interpretation is that m is small enough that θ learns all modes within the first few responses, and the curve shape provides no useful mode count estimate.

A.5 Non-Uniform Mode Weights

For non-uniform weights $\{w_i\}_{i=1}^m$, the expected probability that draw $k+1$ hits a new mode is:

$$p_{\text{new}}(k) = \sum_{i=1}^{m} w_i \,(1 - w_i)^k \tag{36}$$

This no longer has a clean closed form for k^*. However, the scaling is governed by the effective mode count. For weights that are not too extreme, the crossover occurs at roughly:

$$k^* \approx \frac{\ln 2}{\sum_i w_i^2} \tag{37}$$

where $1/\sum_i w_i^2$ is the inverse Simpson index (a standard measure of effective diversity). When weights are uniform, $\sum_i w_i^2 = 1/m$ and this recovers $k^* = m \ln 2$. When weights are concentrated on a few dominant modes, k^* is smaller because the dominant modes are observed early.

A.6 Caveats

1. The linear scaling $k_0 \propto m$ is robust (it follows from the combinatorial structure of the coupon collector process). The constant of proportionality $\ln 2$ is a hypothesis to be tested empirically.

2. The argument assumes that the sigmoid is a good fit to the $\bar{a}_k$ curve. When m is small ($m \ll n$), the curve may be better described by exponential decay (the sigmoid with $k_0 < 1$), and the inflection-point interpretation does not apply.

3. For non-uniform weights, the inverse Simpson index approximation is heuristic. Modes with very small weight $w_i \lesssim 1/n$ may never appear in the sample and do not contribute to k^* or k_0.

References

James P. Crutchfield and David P. Feldman. Regularities unseen, randomness observed: The entropy convergence hierarchy. *Chaos*, 15:25–54, 2003.

Viet Dac Lai, Chien Van Nguyen, Nghia Trung Ngo, Thuat Nguyen, Franck Dernoncourt, Ryan A. Rossi, and Thien Huu Nguyen. k-LLMmeans: Combining LLMs and k-means for zero-shot clustering of text. *arXiv preprint*, 2024.

Hugh Zhang, Daniel Duckworth, Daphne Ippolito, and Arvind Neelakantan. Trading off diversity and quality in natural language generation. *arXiv preprint*, 2024.